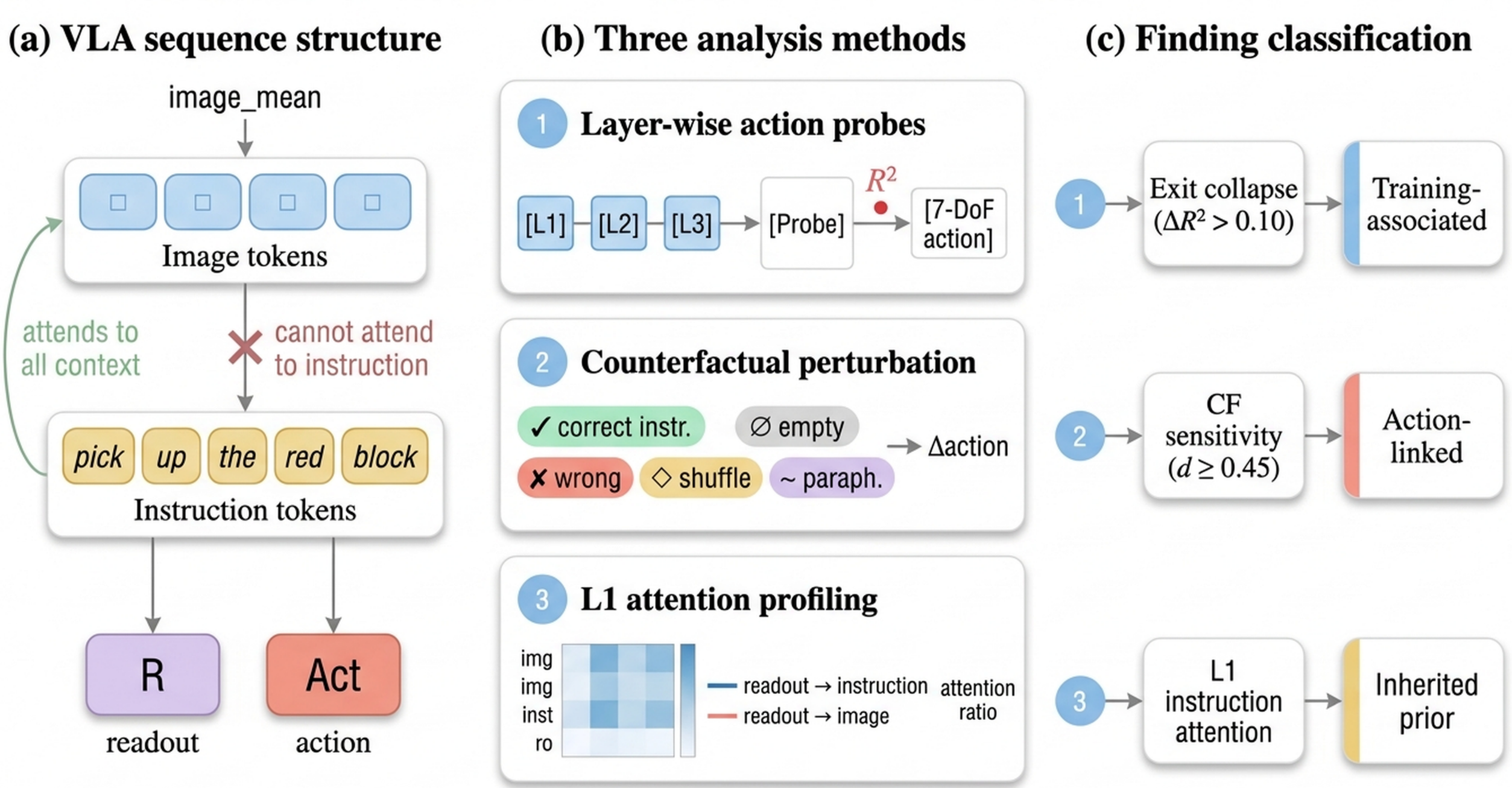

(a) VLA sequence structure
image_mean
Image tokens
attends to all context
cannot attend to instruction
pick
up
the
red
block
Instruction tokens
R
Act
readout
action
(b) Three analysis methods
1
Layer-wise action probes
[L1]
[L2]
[L3]
[Probe]
R^2
[7-DoF action]
2
Counterfactual perturbation
✓ correct instr.
∅ empty
✘ wrong
◇ shuffle
~ paraph.
→ Δaction
3
L1 attention profiling
img
img
inst
ro
readout → instruction
readout → image
attention ratio
(c) Finding classification
1
Exit collapse ($\Delta R^2 > 0.10$)
Training-associated
2
CF sensitivity ($d \geq 0.45$)
Action-linked
3
L1 instruction attention
Inherited prior